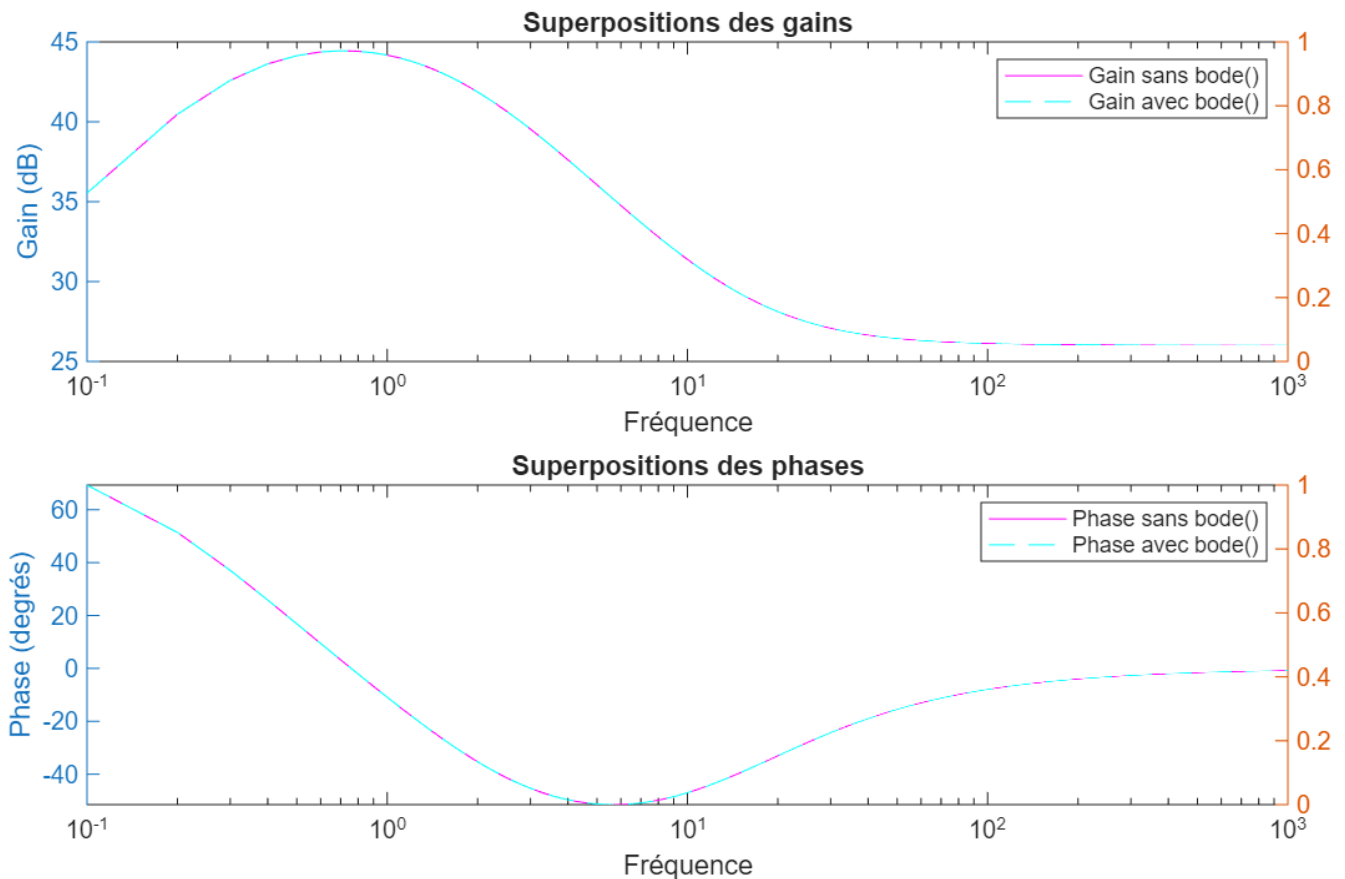


Exercice 1



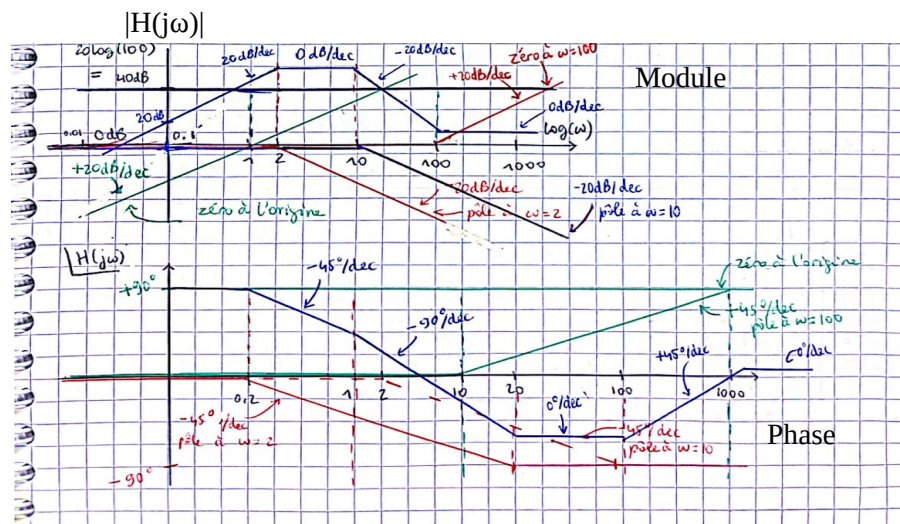
Code correspondant :

```
% Calcul de H sans la fonction bode()
f = 0.1:0.1:1000; % Fréquences de 0.1 à 1000
s = 1j*2*pi*f;
H1 = (20 * s .* (s + 100)) ./ ((s + 2) .* (s + 10)); % Calcul de H(s)

gain1 = 20 * log10(abs(H1)); % Gain en dB
phase1 = angle(H1) * (180/pi); % Phase en degrés

% Calcul de H avec la fonction bode()
H = tf([20 2000 0], [1 12 20]);
[~, ~, wout] = bode(H, 2 * pi * f);
gain2 = squeeze(gain1);
phase2 = squeeze(phase1);
```

Version manuscrite :



Exercice 2

$$1) X(s) = \frac{10(s+1)}{s^2+4s+3} = \frac{10(s+1)}{(s+1)(s+3)} = \frac{A}{(s+1)} + \frac{B}{(s+3)}$$

Trouvons A: $\frac{10(s+1)(s+1)}{(s+1)(s+3)} = \frac{A(s+1)}{(s+1)} + \frac{B(s+1)}{(s+3)}$

Preons $s = -1$: $\frac{10(-1+1)}{(-1+3)} = A + \frac{B(-1+1)}{(-1+3)}$

$$0 = A + 0, \text{ donc, } \boxed{A = 0}$$

Trouvons B:

$$\frac{10(s+1)(s+3)}{(s+1)(s+3)} = \frac{A(s+3)}{(s+1)} + \frac{B(s+3)}{(s+3)}$$

Preons $s = -3$: $\frac{10(-3+1)}{(-3+1)} = \frac{A(-3+3)}{(s+1)} + B$

$$\frac{-20}{-2} = 10 = 0 + B, \text{ donc, } \boxed{B = 10}$$

Ainsi, $\frac{10(s+1)}{s^2+4s+3} = \frac{10}{s+3} = X(s) \xrightarrow{\mathcal{L}^{-1}} \boxed{x(t) = 10e^{-3t}u(t)}$

$$2) X(s) = \frac{10(s+1)e^{-2s}}{s^2+4s+3} = \frac{10(s+1)e^{-2s}}{(s+1)(s+3)} = \left(\frac{A}{(s+1)} + \frac{B}{(s+3)} \right) e^{-2s}$$

Trouvons A: $\frac{10(s+1)(s+1)}{(s+1)(s+3)} = \frac{A(s+1)}{(s+1)} + \frac{B(s+1)}{(s+3)}$

Preons $s = -1$: $\frac{10(-1+1)}{(-1+3)} = A + \frac{B(-1+1)}{(s+3)}$

Trouvons B: $0 = A + 0, \text{ donc } \boxed{A = 0}$

$$\frac{10(s+1)(s+3)}{(s+1)(s+3)} = \frac{A(s+3)}{(s+1)} + \frac{B(s+3)}{(s+3)}$$

Preons $s = -3$: $\frac{10(-3+1)}{(-3+1)} = \frac{A(-3+3)}{(-3+1)} + B$

$$\frac{-20}{-2} = 10 = 0 + B, \text{ donc } \boxed{B = 10}$$

Ainsi, $X(S) = \frac{10(S+1)e^{-2S}}{S^2+4S+3}$

$\xrightarrow{\mathcal{L}^{-1}} \boxed{x(t) = 10 e^{-3(t-2)} u(t-2)}$

3) $X(S) = \frac{20}{S(S^2+10S+16)} = \frac{20}{S(S+2)(S+8)} = \frac{A}{S} + \frac{B}{(S+2)} + \frac{C}{(S+8)}$

Trouvons A : $\frac{20S}{S(S+2)(S+8)} = \frac{A \times S}{S} + \frac{B \times S}{(S+2)} + \frac{C \times S}{(S+8)}$

Prenons $S=0$: $\frac{20}{2 \times 8} = A + 0 + 0$

$\frac{20}{16} = \frac{5}{4} = A$, donc $\boxed{A = \frac{5}{4}}$

Trouvons B : $\frac{20(S+2)}{S(S+2)(S+8)} = \frac{A(S+2)}{S} + \frac{B(S+2)}{(S+2)} + \frac{C(S+2)}{(S+8)}$

Prenons $S=-2$: $\frac{20}{-2(-2+8)} = \frac{A(-2+2)}{-2} + B + \frac{C(-2+2)}{(-2+8)}$

$\frac{20}{-12} = 0 + B + 0$, donc $\boxed{B = -\frac{5}{3}}$

Trouvons C : $\frac{20(S+8)}{S(S+2)(S+8)} = \frac{A(S+8)}{S} + \frac{B(S+8)}{(S+2)} + \frac{C(S+8)}{(S+8)}$

Prenons $S=-8$: $\frac{20}{-8(-8+2)} = \frac{A(-8+8)}{-8} + \frac{B(-8+8)}{(-8+2)} + C$

$\frac{20}{48} = \frac{5}{12} = 0 + 0 + C$, donc $\boxed{C = \frac{5}{12}}$

Ainsi, $X(S) = \frac{20}{S(S^2+10S+16)} = \frac{5}{4S} - \frac{5}{3(S+2)} + \frac{5}{12(S+8)}$

$\xrightarrow{\mathcal{L}^{-1}} \boxed{x(t) = \frac{5}{4} u(t) - \frac{5}{3} e^{-2t} u(t) + \frac{5}{12} e^{-8t} u(t)}$